Swarnavo Mukherjee

Distributed Systems | HPC | LLM Inference | CUDA | GPU Systems

Kolkata, India | +91 9903619331 | swarnavomukherjee03@gmail.com
swarnavo.com | https://www.linkedin.com/in/swarnavo-mukherjee-9a1192263/ | https://github.com/techmonk000

SUMMARY

High-performance **Systems Software Engineer** specializing in **Distributed Systems, LLM Inference infrastructure**, and GPU acceleration. Strong foundation in **data structures, algorithms**, **object-oriented programming**, and system design. Experienced in Python, Java, cloud infrastructure, and full-stack development. Proficient in **CUDA, Rust, and C++** to optimize memory management and scale bare-metal applications. Proven track record of developing custom x86_64 kernels, high-throughput backend pipelines, and **infrastructure** which drove business revenue growth up to **$10K MRR**.

TECHNICAL SKILLS

- **Languages:** Rust, C++, CUDA, Python, Java, JavaScript, TypeScript, PTX inline ASM.
- **Systems & GPU Infrastructure:** Async runtimes (Tokio), Flash Attention, Paged Attention Kernels, WMMA Tensor-Core Programming, Tensor-Core GEMM, VMM, x86_64 Architecture, Memory Virtualization.
- **Backend & Cloud Architecture:** AXUM, Tokio, Tower, Node.js, REST APIs, Next.js, PostgreSQL, Docker, AWS EC2, AWS S3.
- **Machine Learning & Research:** PyTorch, Safe-tensors, KV Cache Compression, Quantization (FP8/INT4), Deep Learning Classifiers.
- **Core CS:** Data Structures, Algorithms, OOP, Operating Systems, DBMS.

PROFESSIONAL EXPERIENCE

Alliance Australia Property Pvt. Ltd | Software Engineer (Full Time) | *05/2025 – 04/2026*

- **Delivery Results:** Engineered core internal LLM-driven infrastructure for automated property valuation reports, reducing document generation turnaround time by 60%.
- **Customer Obsession:** Developed high-throughput data ingestion pipelines and map-based analytics, directly scaling company lead generation to 100+ active leads and accelerating revenue to $10K MRR
- **Insist on the Highest Standards:** Architected an AI-based x86_64 kernel that proactively detected 4+ critical vulnerabilities, improving end-to-end system security and data integrity by 40%.

TECHNICAL PROJECTS

Nebion — GPU Hypervisor Engine (Rust, CUDA) | (GitHub Link)

- **Developed** a multi-tenant LLM inference engine from scratch featuring handwritten CUDA kernels (GEMM, attention, paged KV decode) to successfully deploy Qwen-70B on a single NVIDIA H100 GPU.
- **Designed** a 4-tier adaptive quantization fabric (FP16/FP8/INT8/INT4) leveraging an attention-mass EMA solver to execute real-time per-layer precision swapping, bypassing physical hardware limitations.
- **Built** a 3-tier KV cache system mapping HBM to pinned host RAM to NVME, expanding concurrent-user capacity by 4x on memory-bound servers.
- **Implemented** an SLA-aware scheduling loop with mid-flight P0–P3 pre-emption, yielding a 3x to 4x increase in concurrent user volume during production spikes on Qwen-7B.

Cyber-OS — AI-Kernel Security Subsystem (Rust, WHPX) | (GitHub Link)

- **Engineered** a complete **ring-0 systems** stack featuring a Rust WHPX Virtual Machine Monitor (VMM) and a 5K-line no-std x86_64 kernel, achieving sub-microsecond interrupt handling at 100 Hz.
- **Built** an in-kernel gradient-boosted malware classifier integrating MalConv2 CNN and Light-GBM-ABI drop-ins, reducing false-positive rates by 96% across a 22,000-file target environment.
- **Deployed** an interactive Rust shell utilizing persistent SHA-256 validation, automated quarantines, and a 5-tier verdict system to create an end-to-end threat triage loop.

EDUCATION

University of Engineering & Management | Kolkata, India | Graduation Year - 2026

- Bachelor of Technology (B.Tech.) in Computer Science and Technology